

■ MASTER DOSSIER: PROJECT A.R.T.E.

Autonomous Regenerative Terra-Engine

Version A (Low-Cost Robust) + Version B (High-Performance Upgrade)

Status: 2026

0. Executive Summary

A.R.T.E. is an autonomous soil healing and energy supply system that:

- Regenerates soils
- Injects water and nutrients directly into the root zone
- Produces pellets from local biomass
- Operates with recycled aluminum as an energy source
- Functions completely without diesel, hydraulics, or external infrastructure

It consists of four core modules:

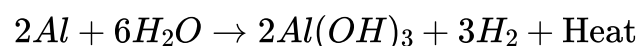
- Metal-Fuel Reactor (1–3 kW, 48V DC)
- A.R.T.E. Forge (Pellet Press)
- A.R.T.E. 2.0 Vehicle Platform
- Terra-Engine (Vibration + Injection)

All modules are available in Version A (Low-Cost) and Version B (High-Performance).

1. ENERGY SYSTEM: METAL-FUEL CORE

1.1 Operating Principle

Reaction:



The reaction produces:

- Heat → Power generation
- Hydrogen → Fuel cell or micro turbine
- Waste: Aluminum hydroxide (recyclable)

1.2 Performance Data

Parameter	Value
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Continuous output	1–3 kW
Voltage	48V DC
Runtime per filling	6–12 h
Aluminum consumption	5–8 kg
Water consumption	6–10 L
Waste	2–3 L Al(OH) ₃

1.3 Dimensions

- Height: 45 cm
- Diameter: 28 cm
- Wall thickness: 3–4 mm stainless steel
- Filler opening: 12–15 cm

1.4 Version A – Low-Cost

- Thermoelectric Generators (TEG)
- Mechanical water supply
- Passive cooling system
- Mechanical pressure relief valve

1.5 Version B – High-Performance

- Micro steam turbine or PEM fuel cell
- Electronically regulated water supply
- Liquid cooling
- Digital sensors (pressure, temperature, gas flow)

1.6 Formulas

Energy per kg aluminum:

$$E = 8.1 \text{ kWh/kg}$$

At 5–8 kg:

40.5–64.8 kWh

Water requirement:

1 kg Al → 1.2 L Water

2. PELLET PRESS: A.R.T.E. FORGE

2.1 Dimensions

- Height: 1.80 m
- Width: 0.90 m
- Depth: 1.30 m

2.2 Core Components

- 400 mm flat die
- 2 rollers (120–150 mm)
- 48V motor (1–2 kW)
- Gearbox 1:20
- Hopper with agitator
- Drying zone (waste heat 60–120 °C)

2.3 Pellet Recipe

Typical mixture:

- 70–80 % Biomass
- 5–10 % Binder (starch/clay)
- 2–5 % Hydrogel
- 5–10 % Nutrient concentrate
- Remainder water

2.4 Throughput

20–50 kg/h

2.5 Version A – Low-Cost

- Fixed speed
- Mechanical backstop
- Passively dried

2.6 Version B – High-Performance

- Speed control
- Humidity sensors
- Airflow control
- Data logging

3. VEHICLE PLATFORM: A.R.T.E. 2.0

3.1 Dimensions

- Length: 1.80–2.00 m
- Width: 1.20–1.40 m
- Height: 1.00–1.20 m
- Weight: < 300 kg

3.2 Drive

- 4× 48V hub motors (500–800 W)
- Differential steering
- Low-pressure tires (20–25 cm wide)

3.3 Gradeability

25°

3.4 Center of Gravity

30–40 cm above ground

3.5 Version A – Low-Cost

- Steel frame (40×40×3 mm)
- Rigid axles
- Simple electronics

3.6 Version B – High-Performance

- Partial suspension
- FEM-optimized frame
- Larger battery (48V, 40–60 Ah)

4. TERRA-ENGINE: SOIL HEALING

4.1 Dimensions & Structure

- Working width: 1.20 m
- 3 spikes (HRC 55–60)
- Spike length: 30–40 cm
- Spike spacing: 0.2 m / 0.6 m / 1.0 m

4.2 Vibration

- Frequency: 40–80 Hz
- Amplitude: 2–4 mm

4.3 Injection

- Pressure: 2–4 bar
- Flow rate: 0.2–0.5 L/min per nozzle
- Lines: EPDM/PU
- Filter: 100–200 μm

4.4 Version A – Low-Cost

- Vibration motors (eccentric)
- Pump ON/OFF
- Fixed flow rate

4.5 Version B – High-Performance

- Regulated vibration
- Flow rate sensors
- Pressure control
- Adaptation to driving speed

5. SOFTWARE ECOSYSTEM

5.1 Architecture

- Hardware abstraction
- Communication layer
- Logic layer
- Mission manager
- Telemetry

5.2 Version A – Low-Cost

- ESP32/STM32
- No ROS
- GPS + IMU + Lidar stop
- Terra-Engine ON/OFF
- Reactor monitoring

5.3 Version B – High-Performance

- NVIDIA Jetson
- ROS2
- GPS-RTK
- Lidar mapping
- AI modules (optional)
- Complete dashboard

6. SAFETY & CERTIFICATION

6.1 Relevant Standards

- Machinery Directive 2006/42/EC
- Low Voltage Directive
- EMC Directive
- Pressure Equipment Directive
- ISO 18497 (Autonomous agricultural machinery)
- ISO 25119 (Safety in agriculture)

6.2 Safety Measures

- Emergency stop
- Protective grid
- Pressure relief valve
- Temperature monitoring
- Watchdog
- Logging

7. PRODUCTION & MARKET LAUNCH

7.1 Version A – Low-Cost

- Local manufacturing
- Standard parts
- Minimal electronics

7.2 Version B – High-Performance

- CNC manufacturing
- Precision sensors
- Data logging

7.3 Pilot Strategy

- Phase 1: Test field
- Phase 2: NGO pilot
- Phase 3: Scaling

8. ECOLOGICAL IMPACT

8.1 CO₂ Savings

- Diesel replacement: 50 kg CO₂/ha
- 100 ha/year → 5 tons CO₂ per bot
- 5 bots → 25 tons CO₂/year

8.2 Humus Buildup

- Long-term carbon sequestration
- Improvement of soil structure
- Reduction of erosion

9. COMPLETE FORMULA COLLECTION

Energy Aluminum:

$$E = 8.1 \text{ kWh/kg}$$

Water requirement:

$$W = 1.2 \text{ L/kg Al}$$

Application rate:

$$Q = A \cdot D$$

with

$$A = \text{Area per second}$$

$$D = \text{Dosage per } m^2$$

Tilt angle:

$$\tan(\alpha) = \frac{\text{half track width}}{\text{center of gravity}}$$

10. CLOSING WORDS

A.R.T.E. is a complete, realistically buildable, scalable system that:

- Regenerates soils
- Provides energy

- Saves water
- Sequesters CO₂
- Creates local value
- Is independent of fossil fuels

This master dossier forms the basis for:

- Production
- Certification
- Pilot projects
- Investors
- NGOs
- Universities
- Government programs